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**Reproducing and Extending a Hybrid Ensemble Model for
APT Attack Detection: How Much of the Reported Performance
Survives?**

Interim Project Report
Research in Information Security

Replication of:

*Saini, Bhat Kasaragod, Prakasha & Das (2023). "A hybrid ensemble machine learning model for detecting APT attacks based on network behavior anomaly detection."
Concurrency and Computation: Practice and Experience, 35(28), e7865.*

Submitted by

Name	Roll Number
Achyutananda Sahoo	[roll number]
[team member]	[roll number]
[team member]	[roll number]

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Abstract

Saini et al. (2023) propose a “hybrid” ensemble of Random Forest (RF) and XGBoost, combined by averaging predicted probabilities, for detecting Advanced Persistent Threat (APT) traffic, and report accuracies of 98.92%, 99.91%, 99.24% and 97.11% on CSE-CIC-IDS2018, CIC-IDS2017, NSL-KDD and UNSW-NB15. We reproduce the full pipeline (cleaning, per-file stratified sampling, Pearson and mutual-information feature selection, SMOTE-Tomek balancing, five baselines and the hybrid) on all four datasets, and add two extensions the paper names as future work: a *stacked* hybrid in which a trained meta-learner combines RF, XGBoost and an LSTM, and an FT-Transformer as a modern tabular deep model.

On three of the four datasets our reproduction lands within 0.36 accuracy points of the paper, with a lower false-positive rate on all three. UNSW-NB15 is the exception (91.33% vs. 97.11%); keeping the 40% duplicate rows that we remove closes about half of that gap. We also find that the hybrid’s advantage over plain XGBoost is negligible (on CIC-IDS2017 it is a single test flow out of 14,771), that the stacked hybrid makes *exactly* the same predictions as simple averaging on CIC-IDS2017 because the meta-learner ignores the LSTM, and that the FT-Transformer trails the tree ensembles by 0.2–0.9 points while giving the lowest false-positive rate on UNSW-NB15. While reproducing we reverse-engineered an undocumented 2% sampling step for CIC-IDS2017 and documented several internal inconsistencies in the paper. We close with a roadmap for a leakage-free evaluation that tests how much of the reported performance survives.

1 Introduction

1.1 Background

An Advanced Persistent Threat (APT) is a long-running, targeted intrusion: attackers gain a foothold, move laterally and exfiltrate data slowly while blending in with normal traffic. Signature-based defences such as antivirus, firewalls and rule-based IDSs miss such activity because it rarely matches a known pattern. Network behaviour anomaly detection (NBAD) instead learns what normal traffic looks like and flags deviations, and machine learning is the usual tool for learning that boundary from labelled flow records.

1.2 The paper being reproduced

Saini et al. [1] evaluate ten classical classifiers and three deep models on four public intrusion datasets, and propose a hybrid of RF and XGBoost whose predicted probabilities are averaged using the `SimpleClassifierAggregator` of the `combo` library. Their pipeline merges all attack types into a single “anomaly” class (binary classification), balances the classes with SMOTE-Tomek, selects features with Pearson correlation followed by mutual information, and explains the model with SHAP. The hybrid reports the highest accuracy on every dataset. The paper’s own Conclusion names combining machine-learning and deep-learning models, including LSTMs, as future work.

1.3 Our contributions

1. **A complete, open reproduction** of the paper’s pipeline on all four datasets, run as a single configurable codebase (Section 4).
2. **Reverse-engineered, undocumented details**, most notably a 2% stratified sample of CIC-IDS2017 that the paper never mentions but that its own class counts reveal exactly (Section 4.4).
3. **An explanation of the one large gap** (UNSW-NB15) through a duplicate-row ablation (Section 5.2).
4. **Two extensions the paper proposes but does not evaluate**: a stacked ensemble of RF, XGBoost and an LSTM with a trained meta-learner, and an FT-Transformer (Section 4.6).
5. **A critical reading** of the paper’s evaluation, listing verifiable internal inconsistencies and a roadmap for a leakage-free re-evaluation (Sections 6.4 and 8).

2 Related Work

- **Ensembles for intrusion detection.** RF and gradient-boosted trees are consistently strong on flow-feature datasets such as CIC-IDS2017 and CSE-CIC-IDS2018; the paper’s comparison table cites D’hooge et al. [2], who also showed that models trained on one CIC dataset transfer poorly to the other.
- **Deep learning on tabular data.** Gorishniy et al. [3] introduced the FT-Transformer, which turns each feature into a token and applies self-attention; Grinsztajn et al. [4] showed that tree ensembles still tend to outperform deep models on medium-sized tabular data.
- **Dataset quality.** Engelen et al. [5] documented labelling and flow-construction errors in CIC-IDS2017, which matter when accuracies are reported to two decimal places.
- **Stacking.** Stacked generalisation [6] trains a meta-learner on out-of-fold base-model predictions and is the standard alternative to fixed-weight averaging.

3 Datasets and Exploratory Analysis

All four datasets from the paper were used. Before building the cleaning step we ran an exploratory analysis of the *raw* files (one notebook per dataset) to decide, from evidence, what the cleaning must handle.

Table 1: Datasets as used in this study. “Used rows” is after sampling and duplicate removal.

Dataset	Raw files	Raw rows	Raw attack share	Used rows	Sampling
NSL-KDD	2	148,517	48.1%	147,888	none
UNSW-NB15	2	257,673	63.9%	154,098	none
CIC-IDS2017	8	≈2.83M	≈19.7%	49,235	2% per file
CSE-CIC-IDS2018	10	16.23M	16.9%	30,838	0.2% per file

3.1 Findings that shaped the cleaning step

- **Class imbalance differs sharply.** NSL-KDD and UNSW-NB15 are close to balanced (48% and 64% attack), whereas the two CIC datasets are 17–20% attack, so SMOTE-Tomek matters most for them (Figure 1).
- **CSE-CIC-IDS2018:** the header row is repeated 59 times inside the data (these rows would otherwise be labelled as attacks); **Flow Byts/s** and **Flow Pkts/s** contain the literal text **Infinity**; and one 3.9 GB file (*02-20-2018.csv*) carries four extra identifier columns (**Flow ID**, **Src IP**, **Src Port**, **Dst IP**) that the paper also drops.
- **CIC-IDS2017:** column names carry leading spaces (**' Label '**); eight columns are constant; the same two rate columns contain infinities; and 65 feature pairs are correlated above 0.9.
- **UNSW-NB15:** **label** is already binary and **attack_cat** names the attack family, so both must be excluded from the features (a leakage trap); 40% of rows are exact duplicates.
- **NSL-KDD:** no header row, three text categoricals (**protocol_type**, **service**, **flag**), one constant column and 13 highly correlated pairs.

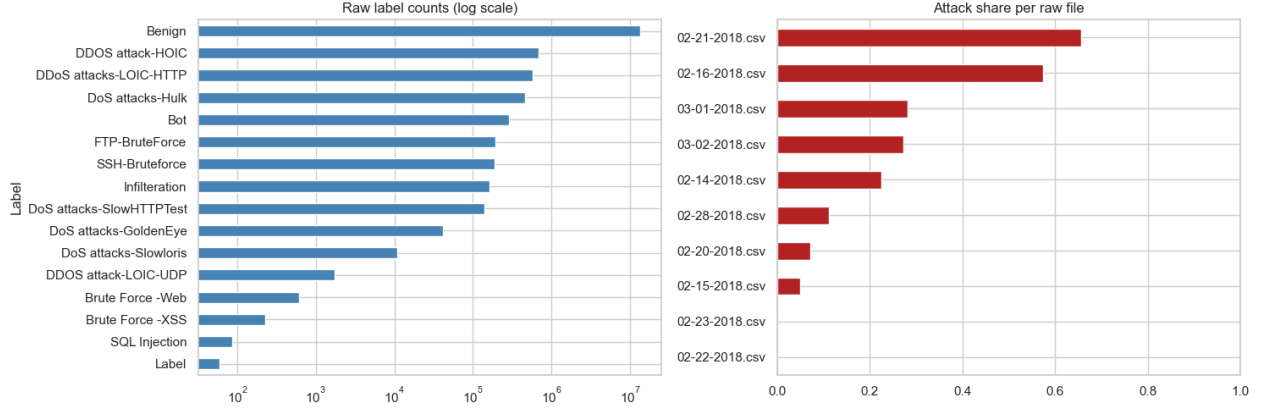


Figure 1: CSE-CIC-IDS2018 raw labels (exact counts over all 16.2M flows). Left: label counts on a log scale; the “Label” bar is the header row repeated inside the data. Right: attack share per daily file, which varies from 0% to 66%. This variation is why the paper samples each file separately, preserving each day’s mix.

4 Methodology

4.1 Pipeline overview

Figure 2 shows the pipeline. Every step is shared by all four datasets; dataset-specific choices (label column, benign value, columns to drop, categoricals, sampling rate) live in a single configuration file.

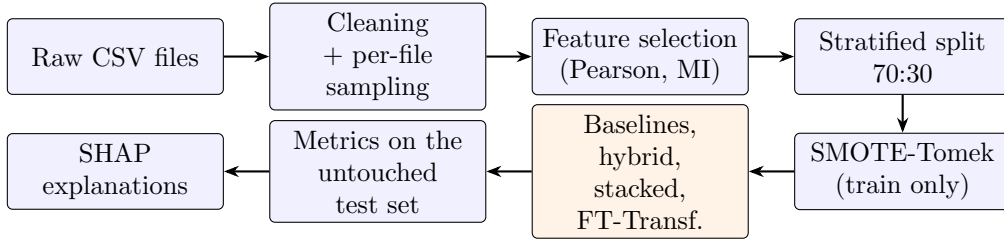


Figure 2: Our pipeline. Unlike the paper, balancing is applied to the training split only, so the test set keeps the natural class distribution.

4.2 Preprocessing

1. **Load and clean labels.** Column names are stripped of whitespace, repeated header rows are removed, and all attack types are collapsed into one class: 0 = benign/normal, 1 = attack, exactly as in the paper.
2. **Per-file stratified sampling.** CSE-CIC-IDS2018 is sampled at 0.2% per file (stated in the paper). CIC-IDS2017 is sampled at 2% per file (*not* stated in the paper, see Section 4.4).
3. **Drop identifiers.** Timestamp is dropped everywhere (as in the paper), plus Flow ID and IP/port identifiers for 2018, and id and attack_cat for UNSW-NB15.
4. **Encode categoricals.** NSL-KDD and UNSW-NB15 text categoricals are label-encoded; the 2018 Protocol column is one-hot encoded, because the paper’s feature list contains Protocol_6 and Protocol_0.
5. **Remove duplicates, coerce to numbers, replace infinities.** Values are converted to numbers *before* infinities are replaced, since the text Infinity only becomes a real infinity on conversion.

4.3 Feature selection

Following the paper: zero-variance features are removed, then for every pair of features correlated above 0.9 one is dropped, then the 30 features with the highest mutual information with the label are kept. Table 2 compares the counts. In that first run on CSE-CIC-IDS2018, 27 of our 30 selected features also appear in the paper’s top 30 (its Figure 8), and the leading features agree: `Init Fwd Win Byts`, `Dst Port`, `Fwd Pkt Len Max` and `Fwd Pkt Len Mean`.

Table 2: Feature counts through selection (non-constant $\rightarrow$ after correlation pruning $\rightarrow$ kept). Our CIC counts come from the first preparation run (CIC-IDS2017 before the Wednesday file was added; CSE-CIC-IDS2018 before `Protocol` was one-hot encoded).

Dataset	Ours	Paper
NSL-KDD	40 $\rightarrow$ 33 $\rightarrow$ 30	not reported
UNSW-NB15	42 $\rightarrow$ 32 $\rightarrow$ 30	not reported
CIC-IDS2017	70 $\rightarrow$ 38 $\rightarrow$ 30	? $\rightarrow$ 36 $\rightarrow$ 30
CSE-CIC-IDS2018	70 $\rightarrow$ 36 $\rightarrow$ 30	71 $\rightarrow$ 40 $\rightarrow$ 30

4.4 Reverse-engineering the CIC-IDS2017 sampling

The paper describes sampling only for CSE-CIC-IDS2018, yet its CIC-IDS2017 class table (Table 4) totals about 56,600 rows, far below the ≈ 2.83 M rows in the dataset. Dividing each of the paper’s class counts by the raw count reveals the step (Table 3): every attack class is exactly 2% of its raw count. The authors therefore took an undocumented 2% stratified sample, and we do the same. This check also showed that one of our files (Wednesday, the DoS day) was initially missing; it was added before the final runs.

Table 3: The paper’s CIC-IDS2017 counts divided by the raw counts.

Class	Raw count	Paper (Table 4)	Ratio
PortScan	158,930	3,179	2.00%
DDoS	128,027	2,561	2.00%
FTP-Patator	7,938	159	2.00%
SSH-Patator	5,897	118	2.00%
Bot	1,966	39	1.98%

4.5 Balancing, splitting and models

Data are split 70:30 with stratification (the paper’s text says 70:30; its pseudocode says 80:20). SMOTE-Tomek is then fitted on the **training split only**. The paper balances *before* splitting, which places synthetic samples interpolated from test rows into training; we avoid that leak deliberately, so our test sets keep their natural class mix.

Reproduction models. Decision Tree, Logistic Regression and MLP (both with feature standardisation, which the paper’s methodology figure includes), Random Forest (200 trees), XGBoost (200 trees), and the paper’s **averaging hybrid**: the mean of RF and XGBoost predicted probabilities, thresholded at 0.5, which is what `combo`’s average aggregator computes. We evaluate a representative five baselines rather than all thirteen of the paper’s models.

4.6 Extensions

Stacked hybrid (RF + XGBoost + LSTM). The paper’s hybrid uses a fixed 50/50 average. Stacked generalisation instead *learns* how to combine models: out-of-fold probabilities from three-fold cross-validation of each base model train a logistic-regression meta-learner, and the base models are then refit on the full training set. The deep base learner is an LSTM (hidden size 32, 12 epochs),

which follows the paper’s future-work suggestion; each flow’s 30 features are fed as a length-30 sequence.

FT-Transformer. Following Gorishniy et al. [3]: each feature gets its own learned linear embedding (a “token”), a [CLS] token is prepended, three pre-norm Transformer blocks (token width 64, 8 heads) mix the tokens, and a head reads the [CLS] output. Inputs are quantile-normalised; training uses AdamW (learning rate 10^{-4}) with early stopping on a 10% validation split, up to 40 epochs.

4.7 Metrics and implementation

Accuracy, precision, recall, F1, false-positive rate (FPR), false-negative rate (FNR) and ROC-AUC from predicted probabilities are reported on the held-out test set. All randomness is seeded (`random_state=42`); two repeated runs produced identical numbers. The code is in Python (scikit-learn, imbalanced-learn, XGBoost, PyTorch, SHAP); experiments ran on Kaggle, partly with an NVIDIA T4 GPU.

5 Results

5.1 Reproduction of the paper’s hybrid

Table 4: The paper’s averaging hybrid: reported vs. reproduced. Values in %, except AUC. The paper’s AUCs are read from its ROC figures (none was transcribed for CSE-CIC-IDS2018).

Dataset	Source	Accuracy	Precision	Recall	F1	FPR	AUC
NSL-KDD	Paper	99.24	99.28	99.09	99.18	0.62	0.992
	Ours	99.60	99.64	99.53	99.59	0.33	0.9999
UNSW-NB15	Paper	97.11	95.89	99.04	97.44	5.29	0.969
	Ours	91.33	89.11	91.67	90.37	8.94	0.980
CSE-CIC-IDS2018	Paper	98.92	99.47	98.35	98.90	0.52	–
	Ours	98.59	97.66	92.59	95.06	0.38	0.976
CIC-IDS2017	Paper	99.91	99.88	99.95	99.91	0.12	0.999
	Ours	99.86	99.60	99.74	99.67	0.10	0.9999

On NSL-KDD, CSE-CIC-IDS2018 and CIC-IDS2017 the reproduced accuracy is within 0.36 points of the paper, and our false-positive rate is lower on all three. Our lower 2018 recall and F1 are expected: our test set keeps the natural 85/15 benign-to-attack mix, whereas the paper’s appears to have been balanced before splitting, which inflates recall. Per-model results for every dataset are in Appendix A; the paper’s model ranking reproduces, with tree ensembles on top and the hybrid best or within 0.01 points of the best model.

5.2 The UNSW-NB15 gap and duplicate rows

UNSW-NB15 is the one dataset where we fall well short (91.33% vs. 97.11%). 40% of its rows are exact duplicates; we remove them, and the paper does not say whether it did. Keeping duplicates lets identical rows sit in both training and test sets:

Table 5: UNSW-NB15 with and without duplicate removal (averaging hybrid).

Setting	Accuracy	F1	FPR
Duplicates removed (ours)	91.33	90.37	8.94
Duplicates kept (ablation)	94.05	95.32	7.25
Paper	97.11	97.44	5.29

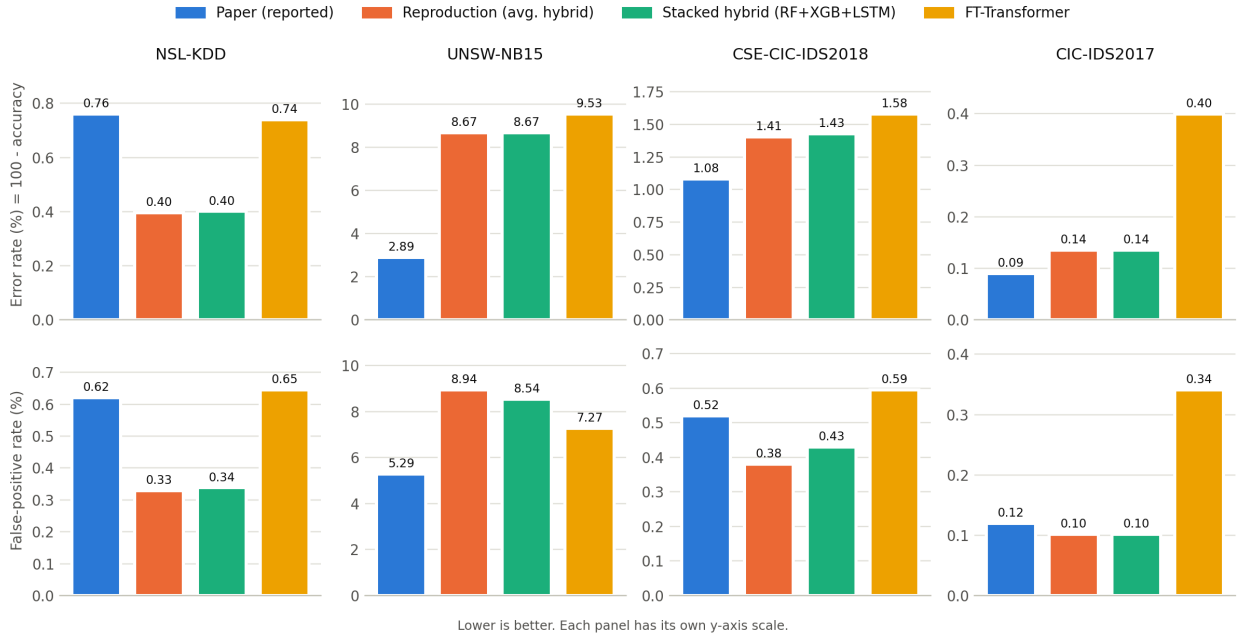
Keeping duplicates recovers 2.7 of the 5.8-point gap. The remainder is plausibly explained by balancing before the split and by the paper’s different, undescribed subset: its Table 6 lists about 81,000 records, while the official partitions contain 257,673.

5.3 Extensions: stacked hybrid and FT-Transformer

All three models below were trained and tested on identical splits, so they compare directly.

Table 6: Our extensions vs. the reproduced averaging hybrid. Accuracy and FPR in %. Best of our three in bold.

Dataset	Averaging hybrid		Stacked (RF+XGB+LSTM)		FT-Transformer	
	Acc.	FPR	Acc.	FPR	Acc.	FPR
NSL-KDD	99.60	0.33	99.60	0.34	99.26	0.65
UNSW-NB15	91.33	8.94	91.33	8.54	90.47	7.27
CSE-CIC-IDS2018	98.59	0.38	98.57	0.43	98.42	0.59
CIC-IDS2017	99.86	0.10	99.86	0.10	99.60	0.34

**Figure 3:** Error rate (100 – accuracy) and false-positive rate for the paper’s reported hybrid, our reproduction, and our two extensions. Lower is better; each panel has its own scale.

6 Discussion

6.1 The hybrid’s advantage is negligible

The paper’s central claim, that the hybrid beats every other model, holds in our runs only by margins that no single-seed experiment can distinguish from noise. On CIC-IDS2017 the hybrid scores

99.865% and XGBoost alone 99.858%: *one* additional correct prediction out of 14,771 test flows. On NSL-KDD, XGBoost alone is marginally ahead (99.61% vs. 99.60%). A claim of superiority needs repeated runs and a significance test, which the paper does not report.

6.2 Stacking collapses to averaging

A trained combiner is strictly more expressive than a fixed average, yet the stacked hybrid never improves on it. On CIC-IDS2017 its predictions are *identical* to the averaging hybrid’s (same accuracy, precision, recall, FPR and FNR to every decimal place); only AUC differs, and it is slightly lower. This indicates that the meta-learner relies almost entirely on RF and XGBoost and gives the LSTM little weight (its weights were not logged in these runs). The LSTM is the weakest base model (training loss 0.12–0.22, unstable on UNSW-NB15). This is expected: stacking gains come from base models that make *different* errors, while RF and XGBoost are highly correlated and the LSTM is weaker than both. There is also a design reason: an LSTM assumes a meaningful order, but the “sequence” here is just the column order of one flow’s features.

6.3 FT-Transformer vs. tree ensembles

The FT-Transformer is 0.2–0.9 points less accurate than the tree ensembles on every dataset, consistent with the finding that trees still lead on medium-sized tabular data [4]. It does, however, give the lowest false-positive rate on UNSW-NB15 (7.27% vs. 8.94%), meaning its errors differ from the trees’. That makes it a better candidate than the LSTM for a stacked ensemble. Its validation loss was still falling at the 40-epoch cap on three datasets, so these numbers are slightly pessimistic.

6.4 Issues found in the paper

While reproducing, we found the following issues, each verifiable from the paper itself:

- **Balancing before splitting.** SMOTE-Tomek is applied to the whole dataset and the result is then split (p. 10), so synthetic samples derived from test rows enter training.
- **Undocumented sampling.** CIC-IDS2017 is sampled at exactly 2% (Table 3), but the text describes sampling only for CSE-CIC-IDS2018.
- **Split ratio.** The text says 70:30; Algorithm 1 says 80:20.
- **Hyperparameters.** Algorithm 1 lists a learning rate of 1.0 and a maximum depth of 4 for the hybrid, which is unusual for XGBoost and would severely limit a Random Forest.
- **Identical deep-learning rows.** The CNN, ANN and MLP results are identical, to two decimals, in the NSL-KDD and UNSW-NB15 tables (Tables 9 and 10), although the datasets differ completely.
- **An impossible F1.** In the UNSW-NB15 table, logistic regression has precision 93.84 and recall 99.66, which implies $F1 \approx 96.66$, yet the table reports 99.66.
- **Hard-label ROC curves.** The ROC curves (e.g. Figures 20, 22 and 24) consist of two straight segments, which is what a curve drawn from hard 0/1 predictions looks like; the reported AUC would then equal balanced accuracy rather than a ranking measure.
- **No variance.** No cross-validation, repeated seeds or significance tests are reported, although many claimed improvements are below 0.3 points.

7 Limitations of This Study

We hold our own work to the same standard:

- **Feature selection sees the whole dataset.** Like the paper, correlation pruning and mutual-information ranking run before the train/test split, which leaks a little test information into feature choice.
- **NSL-KDD split.** We merge the official KDDTrain+ and KDDTest+ files and split randomly (as the paper appears to), instead of using the official test set, which contains attack types absent from training and is known to be much harder.
- **Single seed.** All results come from one seeded split; we report no confidence intervals.
- **Reduced model set.** Five baselines instead of the paper’s thirteen models.
- **No real APT data.** As the paper itself concedes, none of the four datasets contains actual multi-stage APT campaigns; they are flow-level attack/benign captures.
- **Deep-model training budget.** The FT-Transformer was capped at 40 epochs, and some runs executed on CPU.

8 Conclusion and Future Work

We reproduced Saini et al.’s hybrid RF+XGBoost APT detector on all four datasets. The reported performance largely reproduces on NSL-KDD, CSE-CIC-IDS2018 and CIC-IDS2017, while UNSW-NB15 falls short by 5.8 points, about half of which duplicate rows explain. The hybrid’s advantage over a single XGBoost model is negligible, a trained stacking meta-learner collapses to the same predictions, and an FT-Transformer does not beat the trees. The pattern suggests that the 97–99.9% accuracies owe more to the datasets and the evaluation protocol than to the ensemble design.

The next phase will test that directly: *how much of the reported performance survives a rigorous evaluation?*

1. **Leakage-free protocol.** Balancing and feature selection fitted inside each training fold; the official NSL-KDD test sets (KDDTest+, KDDTest-21); a measurement of how much the paper’s balance-then-split order inflates results.
2. **Statistical rigour.** Multiple seeds with stratified cross-validation, mean $\pm$ standard deviation, and paired tests (McNemar, Wilcoxon) against the hybrid.
3. **Generalisation.** Train on CIC-IDS2017 and test on CSE-CIC-IDS2018 (shared CICFlowMeter features); leave-one-attack-out tests as a stand-in for unseen (zero-day) techniques; temporal splits across capture days.
4. **Shortcut features.** Ablate `Dst Port` and `Init Fwd Win Byts`, the top SHAP features, which may encode how the testbed was built rather than attack behaviour; rerun on the corrected CIC datasets [5].
5. **Stronger and more diverse models.** LightGBM and CatBoost, other tabular deep models, and stacking with the FT-Transformer as the diverse base learner.
6. **A real APT angle.** Attack-stage datasets such as DAPT2020, where per-host flow sequences give sequence models (LSTM, Transformer) a meaningful order to learn from.

References

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- [4] L. Grinsztajn, E. Oyallon, and G. Varoquaux, “Why do tree-based models still outperform deep learning on typical tabular data?” in *NeurIPS Datasets and Benchmarks Track*, 2022.
- [5] G. Engelen, V. Rimmer, and W. Joosen, “Troubleshooting an intrusion detection dataset: the CICIDS2017 case study,” in *IEEE Security and Privacy Workshops (SPW)*, 2021.
- [6] D. H. Wolpert, “Stacked generalization,” *Neural Networks*, 5(2):241–259, 1992.

A Full Reproduction Results

All values in %, except AUC. Each caption lists the paper’s reported accuracies where available (the paper’s 2017/2018 baseline tables were not transcribed).

Table 7: NSL-KDD (paper: DT 98.85, LR 87.39, RF 99.15, XGB 98.90, MLP 97.98, hybrid 99.24).

Model	Accuracy	Precision	Recall	F1	FPR	FNR	AUC
Decision Tree	99.40	99.43	99.32	99.37	0.52	0.68	0.9944
Logistic Regression	93.52	94.71	91.61	93.13	4.71	8.39	0.9810
Random Forest	99.55	99.61	99.45	99.53	0.36	0.55	0.9998
XGBoost	99.61	99.64	99.55	99.60	0.33	0.45	0.9999
MLP	99.15	98.97	99.27	99.12	0.95	0.73	0.9993
Averaging hybrid	99.60	99.64	99.53	99.59	0.33	0.47	0.9999

Table 8: UNSW-NB15, duplicates removed (paper: DT 96.24, LR 96.18, RF 96.98, XGB 96.96, MLP 97.98, hybrid 97.11).

Model	Accuracy	Precision	Recall	F1	FPR	FNR	AUC
Decision Tree	88.88	86.52	88.76	87.62	11.04	11.24	0.8910
Logistic Regression	84.14	74.63	97.32	84.48	26.38	2.68	0.9377
Random Forest	91.00	88.48	91.63	90.03	9.52	8.37	0.9778
XGBoost	91.24	89.34	91.13	90.23	8.67	8.87	0.9796
MLP	90.12	86.73	91.79	89.19	11.20	8.21	0.9741
Averaging hybrid	91.33	89.11	91.67	90.37	8.94	8.33	0.9800

Table 9: UNSW-NB15, duplicates kept (ablation).

Model	Accuracy	Precision	Recall	F1	FPR	FNR	AUC
Decision Tree	92.50	94.35	93.88	94.11	9.96	6.12	0.9254
Logistic Regression	88.55	91.17	90.88	91.03	15.59	9.12	0.9595
Random Forest	93.80	95.63	94.62	95.12	7.66	5.38	0.9885
XGBoost	93.83	95.75	94.53	95.14	7.42	5.47	0.9890
MLP	92.39	95.01	92.99	93.99	8.65	7.01	0.9843
Averaging hybrid	94.05	95.86	94.79	95.32	7.25	5.21	0.9897

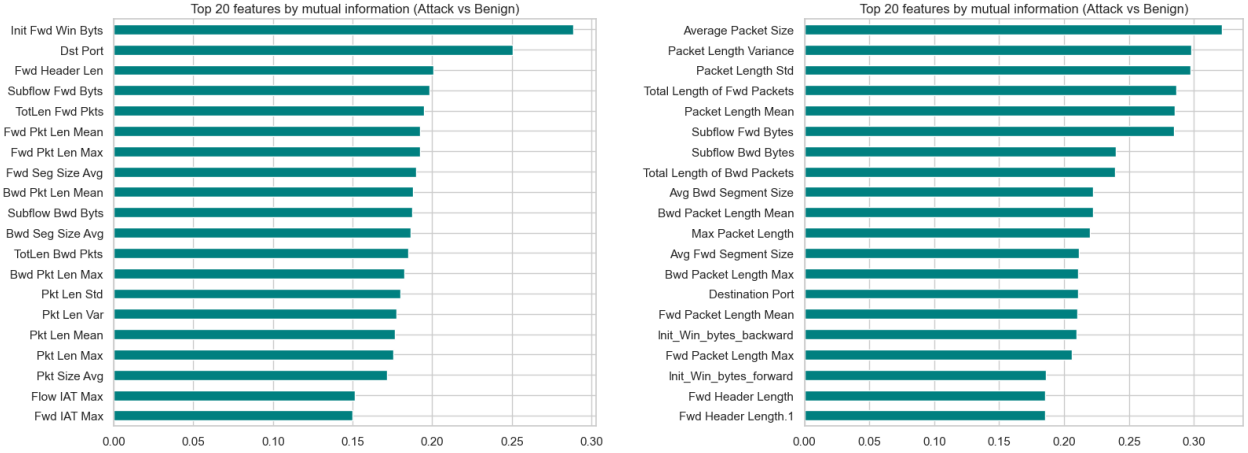
Table 10: CSE-CIC-IDS2018 (paper: RF 98.66, XGB 98.29, hybrid 98.92).

Model	Accuracy	Precision	Recall	F1	FPR	FNR	AUC
Decision Tree	97.65	91.26	92.81	92.03	1.52	7.19	0.9565
Logistic Regression	82.92	45.60	88.30	60.14	18.00	11.70	0.9283
Random Forest	98.57	97.88	92.22	94.97	0.34	7.78	0.9722
XGBoost	98.44	96.74	92.44	94.55	0.53	7.56	0.9755
MLP	98.21	95.05	92.52	93.77	0.82	7.48	0.9727
Averaging hybrid	98.59	97.66	92.59	95.06	0.38	7.41	0.9759

Table 11: CIC-IDS2017, all eight files (paper: RF 99.88, XGB 99.86, hybrid 99.91).

Model	Accuracy	Precision	Recall	F1	FPR	FNR	AUC
Decision Tree	99.62	98.75	99.40	99.08	0.32	0.60	0.9954
Logistic Regression	93.42	77.60	95.40	85.59	7.09	4.60	0.9843
Random Forest	99.80	99.60	99.44	99.52	0.10	0.56	0.9997
XGBoost	99.86	99.54	99.77	99.65	0.12	0.23	1.0000
MLP	99.03	96.10	99.27	97.66	1.04	0.73	0.9986
Averaging hybrid	99.86	99.60	99.74	99.67	0.10	0.26	0.9999

B Exploratory Analysis Figures

**Figure 4:** Top 20 features by mutual information on the raw data. Left: CSE-CIC-IDS2018. Right: CIC-IDS2017 (the first seven files, before the Wednesday file was added).

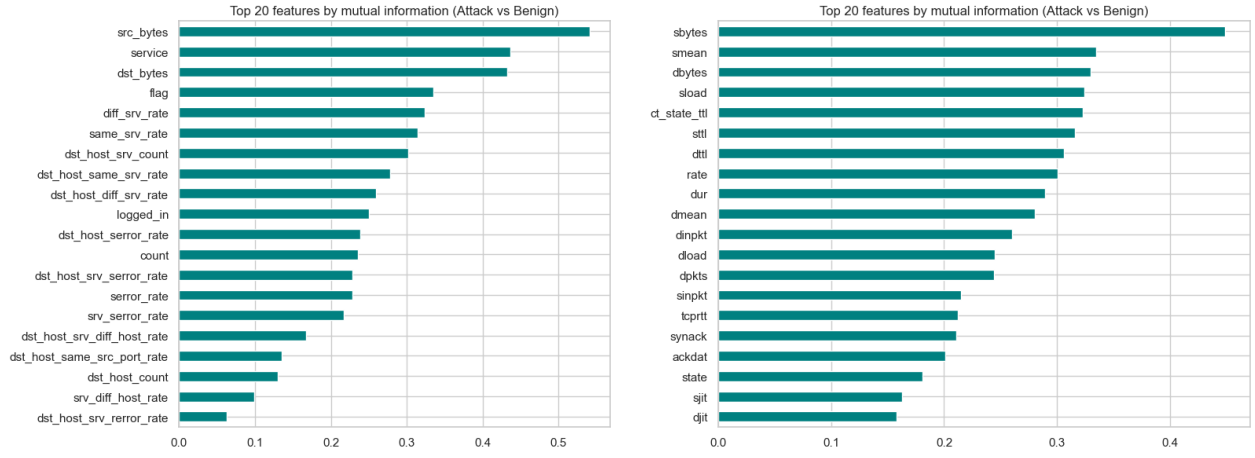


Figure 5: Top 20 features by mutual information. Left: NSL-KDD. Right: UNSW-NB15.